Predicting EV (Electric Vehicle) Energy Consumption using ML (Machine Learning) Techniques

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***Abstract*—Public EV (Electric Vehicle) charging introduces uncertainty in session duration and energy use, posing challenges for distribution networks. Prior work using the ACN-Data (Adaptive Charging Network Data) set shows tree-based models like Random Forest and XGBoost outperform linear and neural models for predicting session energy based on basic time features [1]. However, these models are typically deterministic, lack uncertainty prediction, and are rarely used in grid-aware charging optimization.**

This paper presents a concise, three-layer hybrid model using ACN-Data (Adaptive Charging Network Data) [1]: (1) a stacked ensemble of Random Forest, XGBoost, and K-Nearest Neighbors; (2) a probabilistic forecasting layer with quantile regression neural networks and gradient-boosted trees for energy prediction; and (3) a dynamic optimization module that uses these forecasts to manage charging and control station load within operational limits. The predictive layers rely on core ACN-Data features [1], while the optimization layer additionally uses clearly stated operational assumptions; this keeps the framework reproducible and aligned with prior work.

***Index Terms*—electric vehicles, EV (Electric Vehicle) charging, energy consumption prediction, machine learning, ensemble learning, quantile regression, probabilistic forecasting, optimization, smart charging.**

I. Introduction

Due to the move towards decarbonization in the transportation industry, the central agenda is the replacement of internal combustion engines with electric vehicles. According to recent reviews in Renewable Energy and Energies, EVs (Electric Vehicles) may produce much lower greenhouse gas emissions than internal combustion vehicles, and also, present new problems to the power systems such as higher peak loads, more intermittent demand, and smarter charging control [2,3]. The charging stations in the workplace and especially the ones in the public have a highly fluctuating session-based energy consumption and arrival trends, which is why precise prediction is necessary to make the infrastructure planning and smart charging highly effective.

ACN-Data (Adaptive Charging Network Data) is a major open dataset of workplace EV (Electric Vehicle) charging sessions and a widely used testbed for data-driven charging research [1]. In the work of Rathore et al. [4], models trained on temporal features derived from connection time, done-charging time, and charging duration showed that tree-based methods (Random Forest and XGBoost) outperformed linear regression, ANN (Artificial Neural Network), and DNN (Deep Neural Network) baselines on this tabular task. Their Random Forest model reported an approximately 5.9% relative RMSE (Root Mean Squared Error) improvement and a MAPE (Mean Absolute Percentage Error) of 18.8% (equivalently 0.188 in fractional form) [4].

In research by Khan et al. [5] conducted on multi-model EV prediction showed that models including K-NN (K-Nearest Neighbors), Random Forest, and Extra Trees are capable of predicting a wide range of the performance aspects of EVs (Electric Vehicles) in the city and on the highway with a high R^2 (coefficient of determination) over 0.95 [5]. Hussain et al. [6] also demonstrated on a large real-world charging dataset in Colorado that combination of multiple models, namely, KNN (K-Nearest Neighbors) and RF (Random Forest) with a linear meta-learner could be better than single models with MAE (Mean Absolute Error) = 0.6459 RMSE (Root Mean Squared Error) = 1.7885 and R^2 (coefficient of determination) = 0.9601 to predict energy consumption [6]. Complementary work using Kumar et al. [7] used ensemble stack generalization on KNN, decision trees and Random Forest to predict EV charging behavior (stay duration and energy consumption) on ACN-like data clearly show stacking gains [1,7].

Simultaneously, Zhu et al. [8] created a quantile-based machine learning pipeline to predict trip-level EV (Electric Vehicle) energy consumption in terms of real-world taxi-based data, which they named quantile regression QR (Quantile Regression), quantile regression neural networks QRNN (Quantile Regression Neural Network), quantile XGBoost QEGBR (Quantile Extreme Gradient Boosting Regression), and quantile random forests QRF (Quantile Regression Forest) [8]. Their results show that quantile models capture both point estimates and prediction intervals. Specifically, the global QRNN (Quantile Regression Neural Network) achieved a PMAE (Percentage Mean Absolute Error) of 6.3%, while the online adaptive QRNN reduced prediction error to 5.04% [8].

Lastly, Romano et al. [9] suggested a machine learning-

based EV energy consumption prediction and optimization framework that utilizes a combination of gradient boosting regression, Random Forest, and LSTM (Long Short-Term Memory) in conjunction with an optimization layer that can reassign the driving behavior to reduce the amount of energy use without jeopardizing driver comfort [9]. Their findings indicate that they are saving more than 18% in terms of energy, greater than the energy saving in eco mode settings that are not in use of prediction and optimization in relation to EV energy applications.

Spurred by such developments, this paper poses the following question: Will an ensemble of RF (Random Forest)/XGB (Extreme Gradient Boosting)/KNN (K-Nearest Neighbors) be better able to predict session-level energy predictors of ACN-Data compared to individual RF (Random Forest)/XGB (Extreme Gradient Boosting) [1]? Is it possible to versionize quantile based probabilistic predictions to any ACN session [1] to provide valuable prediction bounds, akin to trip-level energy forecasting of field data? Is it possible to incorporate these deterministic and probabilistic predictions into an easy dynamic optimization model which minimizes peak loads at the ACN stations [1] and meets the user's energy needs? To address these issues, we propose a hybrid structure that merges stacked ensembles, quantile forecasting and optimization into a single ACN-based pipeline [1], and we will evaluate it in Stage-2.

The remainder of this paper is organized as follows. The Proposed Algorithm section presents the overall hybrid framework, including the stacked ensemble, quantile-based probabilistic forecasting layer, and dynamic optimization module. The Experimental Design section describes train-test splits, baselines, and planned ablation studies, while the Datasets section summarizes ACN-Data [1] and preprocessing steps. The Results (Evaluation Metrics) section defines the evaluation criteria and analysis protocol for Stage-2 experiments. Finally, the paper concludes with key takeaways and future directions.

Proposed Algorithm

The framework proposed consists of three major sections: a session based stacked ensemble, deterministic energy prediction, a probabilistic forecasting layer, prediction interval, and a dynamic optimization module, which permits scheduling to be carried out.

A. Overall Hybrid Architecture

As shown in Fig. 1, ACN-Data session logs [1] on the left give the time of connection, done-charging time and delivered energy. These are turned into clean features by a preprocessing and feature-engineering block, such as charging duration, and temporal features (hour of day, day of week, month). These features are then separated into two parallel predictive lines: An ensemble path stacked and creating a point estimate of energy per session. A probabilistic path that generates prediction intervals that are based on quantiles. To the right, an optimization block is performed dynamically, and both the point predictions and intervals are supplied, along with simple operational limits, and the charging decisions are given (e.g. power limits, or closed periods), which minimize peaks at the expense of meeting the energy demands.

This architecture closely follows the model-stacking evidence in [6,7] and the quantile-optimization interplay in [8,9].

B. Session-Level Stacked Ensemble

The session-level ensemble extends the RF (Random Forest)/XGB (Extreme Gradient Boosting) baseline on ACN-Data [1] by adding KNN (K-Nearest Neighbors) and a meta-learner.

Each session is represented by the following base features: charging duration (difference between done-charging and connection timestamps) start hour of day day of week month These features were already effective for RF (Random Forest)/XGB (Extreme Gradient Boosting) in the work of Rathore et al. [4] and are analogous to the temporal indicators used in the Colorado stacking study [6].

Three regressors are trained (Base Models (Level-0)): Random Forest (RF), a strong baseline for ACN session-level prediction [1,4]. XGBoost (XGB), widely successful for tabular EV energy tasks [4,5,6]. K-Nearest Neighbours (KNN), motivated by its high predictive accuracy in the multi-model EV specification study and its role in stacking hybrid ensembles [5,6,7].

Each base model outputs a preliminary energy estimate for each session. Meta-Learner (Level-1). A Linear Regression model takes the three base predictions as input and learns to combine them optimally. This mirrors the stacking hybrid ensemble in [6] where a simple linear meta-model was sufficient to fuse KNN, RF, and Extra Trees outputs [6].

C. Probabilistic Forecasting Layer

In a bid to model any uncertainty, a quantile-based probabilistic layer is incorporated, which is based on the trip-level energy work by Zhu et al. [8]. There are two concrete approaches that are taken into consideration. Quantile Regression Forest (QRF), session level. With the identical features as the stacked ensemble, QRF (Quantile Regression Forest) quantifies conditional quantiles, including the 5th, 50th, and 95th percentiles of the energy of the session. This provides a median and prediction intervals. Session level, quantile XGBoost (QEGBR-like). QEGBR (Quantile Extreme Gradient Boosting Regression)-style models can also be trained to predict upper and lower quantiles by a modification of the XGB (Extreme Gradient Boosting) loss function to a differentiable approximation of quantile loss, a combination of gradient boosting and quantile estimation, [8].

These models together can be used to provide prediction intervals that can be assessed using coverage probability and average width, as in [8].

D. Dynamic Optimization Module

The last layer is a dynamic optimization layer, which employs prediction and interval to control charging decisions. The optimization is developed following the model of Romano et al. that incorporated prediction and control in the process

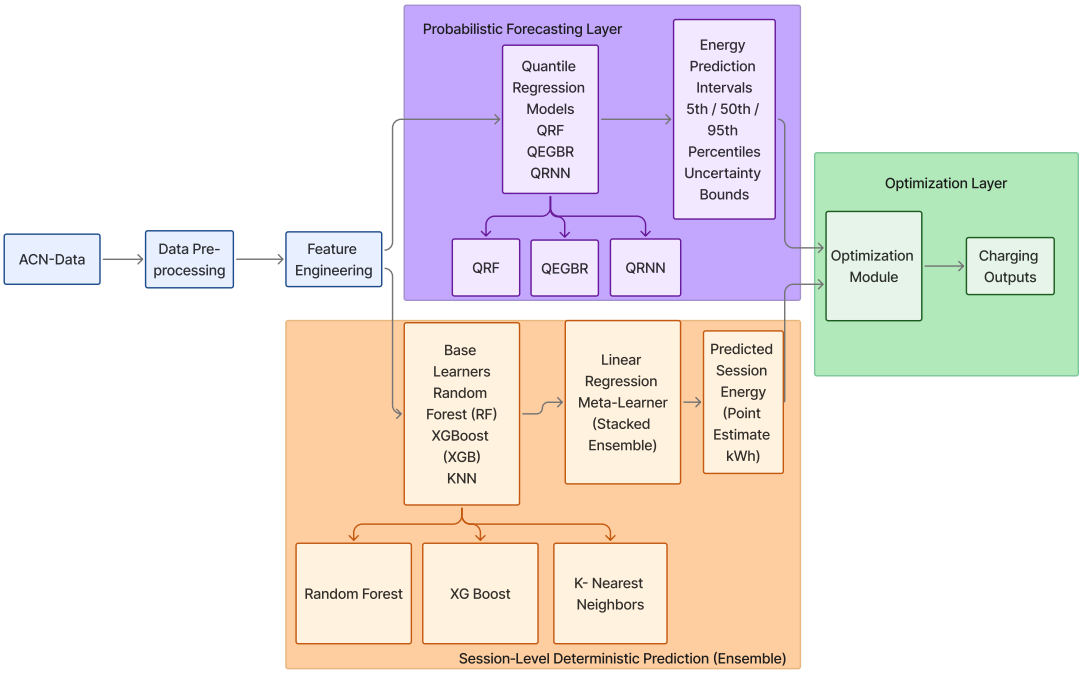

Fig. 1. Overview of the proposed ACN-based hybrid pipeline combining stacked point forecasting, quantile uncertainty estimation, and dynamic charging optimization [1].

of driving energy [9]. The optimization is represented as a constrained problem in the short run.

In a simplified environment, at every step of decision making: Inputs of predicted session energy demand, prediction intervals, station power capacity limits, charger power rating, and user deadlines (e.g., preferred completion times). Objective is to reduce a weighted sum of the forecasted peak load in the station and total energy cost, and punish unsatisfied energy requirements or delayed deliveries. Decision variables signify the amount of power to charge or the time during which the charging may take place during every active session. The amount of powers cannot exceed the capacity of the feeder, each charger must act within its limits, and the amount of required energy (or even a conservative quantile) can be delivered within the remaining time, which is ensured by constraints.

This can be formed as a linear or a mixed- integer program and its purpose in this paper is to demonstrate how the hybrid prediction framework can be straight forwardly used to make grid-aware operational decisions.

II. Additional Inputs and Assumptions

To avoid ambiguity in reproducibility, we separate ACN-native features [1] from optimization-side inputs. ACN-Data [1] provides session timestamps and delivered energy used for prediction. The optimization module additionally requires feeder capacity limit (kW), per-charger power limit (kW), user deadline or preferred completion time (timestamp or slot index), and an optional time-of-use tariff ($/kWh). In Stage-2, these non-ACN inputs will be specified for each experiment as measured values (when available), synthetic scenario values, or fixed defaults; their units and valid ranges will be reported explicitly.

III. Experimental Design

The experimental design follows ACN-Data practice [1] and mirrors evaluation standards in the cited literature.

A. Train-Test Splits

The training and the validation will be done with the use of sessions in October and November 2018, and the sessions of December 2018 will compose a held-out test set, similar to Rathore et al. [4]. Hyperparameter tuning and to produce out-of-fold predictions to train the meta-learner will be calculated during training period with the help of k-fold or rolling-origin cross-validation.

B. Baselines

Baselines at the session level are single RF (Random Forest), XGB (Extreme Gradient Boosting), linear regression, ANN (Artificial Neural Network) and DNN (Deep Neural Network) model [4]. Stronger baselines will be obtained using tuned single RF (Random Forest)/XGB (Extreme Gradient Boosting)/KNN (K-Nearest Neighbors) models, which are optimized, in a fashion akin to [6]. In the case of probabilistic forecasting, QRF (Quantile Regression Forest) and quantile XGB (Extreme Gradient Boosting) will be implemented in comparison with the simpler interval heuristics.

C. Ablation Studies

Planned ablations include: Stacked ensemble vs. optimum single model. QRF (Quantile Regression Forest) vs. QEGBR (Quantile Extreme Gradient Boosting Regression)-style XGB (Extreme Gradient Boosting) coverage and width. Deterministic forecast optimization compared to optimization using conservative quantile based estimates.

These experiments are intended to be executed in Stage-2.

IV. Datasets

Experiments are all grounded upon ACN-data. This data set includes over 30,000 workplace ACN charging sessions, where every session is represented by connector ID (Identifier), connection time, done charging time, energy delivered in kWh and many others [1].

There will be preprocessing followed by: [4] Break down the timestamps of the parses into the standard format. Filter off entries with no timestamps, or with mismatched timestamps. Calculate time variables and hours-days-months. Optional remove the significant outliers whose energy or time can be beyond reasonable operating limits because outliers have been demonstrated to negatively affect model performance in other similar EV energy research [8].

No proprietary or external data is employed so that the framework is not only entirely reproducible.

V. Results (Evaluation Metrics)

There are no empirical results provided in this Stage-1 paper. It, rather, identifies evaluation metrics and analysis procedures to be used in Stage-2.

Point Prediction Metrics: RMSE (Root Mean Squared Error, kWh), MSE (Mean Squared Error, kWh^2), MAE (Mean Absolute Error, kWh), MAPE (Mean Absolute Percentage Error, %), and R^2 (coefficient of determination, unitless) will be used to evaluate session-level models on the test set, as in [4,5,6,8]. If normalized RMSE is additionally reported, the normalization denominator (e.g., mean test-set energy in kWh) will be stated explicitly. These measures enable direct comparison with ACN-based RF (Random Forest)/XGB (Extreme Gradient Boosting) baselines [1] and multi-model EV stacking studies.

- Probabilistic Metrics. The empirical coverage probability of modeled interval with respect to various countless nominal levels (e.g., 90, 95%) and the mean width of the interval will be calculated based on Zhu et al. [8] when operating this model with QRF (Quantile Regression Forest) and QEGBR (Quantile Extreme Gradient Boosting Regression). It is also possible to demonstrate the plots of calibration in which the empirical and nominal coverage are compared to determine the reliability of intervals.
- Optimization Performance. In the case of the dynamic optimization module, the key key performance indicators will include the decrease in the peak station load, alteration in the overall cost of energy (when time-of-use prices are modeled), and the breach of user energy needs. Such metrics will show whether or not the concept of coupling prediction with optimization has practical advantages, similar to those that have been shown to be feasible in [9].

VI. Conclusion

The one-based framework proposed in this paper incorporates three complementary elements, namely: a session-based stacked ensemble, a probabilistic forecasting layer and a dynamic optimization module. The design is grounded in RF (Random Forest)/XGB (Extreme Gradient Boosting) foundations on ACN session-level prediction [1,4], in ensemble-stacking achievements in more general EV data [5,6,7], in quantile-based uncertainty estimation [8], and in ML (Machine Learning)-based optimization structures [9]. Stage-2 experiments will be used to validate performance claims.

The framework brings ACN based research [1] a step closer to the type of integrated, uncertainty aware grid based analytics which is needed to reach a reliable EV charging network by unifying deterministic ensemble prediction, probabilistic forecasting, and operational decision making.

VII. Future Scope

Future directions of this work are to extend feature sets with exogenous information (weather, calendar, partial vehicle information, etc.) when it is available, apply online more adaptive quantile models the way in [8] to make predictions more personalized to frequent users or to a single charger, and to consider more sophisticated optimization formulations such as stochastic programming or reinforcement learning, in the spirit of predictive control in [9]. The second potential direction is to reexamine the station-level modeling with the time-series architectures (e.g., LSTM (Long Short-Term Memory), CNN-LSTM (Convolutional Neural Network–Long Short-Term Memory)) [10,11] as soon as the session-level hybrid architecture is proves to be correct and, therefore, synthesize the two visions to a multi-scale ACN-Data study [1].

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